

ATAI L-Series Heavy Lift Platforms

ATAI L-Series Heavy Lift Aerial Platforms

Engineering Concept Document

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Page 1 — Executive Overview

The ATAI L-Series is a family of large-scale hybrid buoyant aerial platforms designed for ultra-heavy vertical lift, infrastructure logistics, offshore construction, and strategic cargo operations. The system combines helium buoyancy with distributed electric propulsion to create a stable aerial crane capable of lifting loads far beyond conventional helicopters.

Three scalable platforms are defined:

- ATAI L80
- ATAI L120
- ATAI L200

Each platform shares a common engineering philosophy: a rigid disc-shaped buoyant body, distributed propulsion rings, a central structural spine, and a reinforced payload keel supported by mechanical landing legs.

The design enables safe operations in remote environments such as mines, offshore wind farms, desert infrastructure projects, and humanitarian logistics corridors.

Page 2 — Design Philosophy

The ATAI architecture is built around four engineering principles:

1. Hybrid Lift

Lift is produced by both helium buoyancy and active thrust from large distributed fans.

2. Structural Rigidity

Unlike historical airships with flexible envelopes, the ATAI platform uses a rigid internal structure that distributes loads through a central spine.

3. Distributed Propulsion

Multiple large fans arranged in rings provide lift, stability, and directional control.

4. Modular Scalability

The same architecture scales from 80 meters to 200 meters without changing the core design logic.

This allows the product family to target multiple industrial markets while sharing engineering systems.

Page 3 — Global Dimensions

Three main variants exist in the L-Series platform family.

ATAI L80

- Diameter: 80 meters
- Height: ~32 meters

- Primary role: regional logistics and medium heavy lift

ATAI L120

- Diameter: 120 meters
- Height: ~55 meters
- Primary role: offshore energy and infrastructure installation

ATAI L200

- Diameter: 200 meters
- Height: ~100 meters
- Primary role: strategic heavy lift and mega-infrastructure transport

All variants share the same disc-shaped aerodynamic geometry.

Page 4 — Structural Architecture

Each ATAI platform contains three main structural layers.

Helium Lift Body

The upper structure contains multiple helium cells providing passive buoyancy.

Propulsion Deck

The middle layer houses the distributed fan propulsion system.

Payload Keel and Cockpit

The lower section contains the crew capsule, mission systems, and heavy cargo attachment structure.

A central structural spine connects these layers and distributes loads across the entire platform.

Page 5 — Landing System

Each ATAI platform includes a circular landing support ring with 24 mechanical legs.

Each leg contains:

- Industrial compression springs
- Hydraulic shock absorbers
- Wide landing pads

This system stabilizes the vehicle during landing and distributes structural loads evenly across the platform.

Landing Leg Dimensions

L80

- Height: 3 meters
- Width: 6 meters

L120

- Height: 3.5 meters
- Width: 9 meters

L200

- Height: 4 meters
- Width: 12 meters

The legs form a circular support system designed for uneven terrain.

Page 6 — ATAI L80 Specifications

- Diameter: 80 meters
- Height: 32 meters
- Helium Volume: ~200,000 cubic meters
- Helium Lift: ~200 tons

Fan System

- Number of Fans: 16
- Fan Diameter: 12 meters
- Estimated thrust lift per fan: 6–8 tons
- Total Fan Lift: ~110 tons
- Total Lift Capacity: ~310 tons
- Estimated Structural Weight: ~190 tons
- Payload Capacity: 80–100 tons

Primary Markets

- Mining logistics
 - Remote construction
 - Emergency transport
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Page 7 — ATAI L120 Specifications

- Diameter: 120 meters
- Height: 55 meters
- Helium Volume: ~900,000 cubic meters
- Helium Lift: ~900 tons

Fan System

- Number of Fans: 32

- Fan Diameter: 15 meters
- Average Lift per Fan: ~12 tons
- Total Fan Lift: ~380 tons
- Total Lift Capacity: ~1,280 tons
- Estimated Structural Weight: ~650 tons
- Payload Capacity: 500–600 tons

Primary Markets

- Wind turbine installation
 - Oil and gas platforms
 - Large construction logistics
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Page 8 — ATAI L200 Specifications

- Diameter: 200 meters
- Height: 100 meters
- Helium Volume: ~4,000,000 cubic meters
- Helium Lift: ~3,560 tons

Fan System

- Number of Fans: 64
- Fan Diameter: 18 meters
- Average Lift per Fan: ~17.8 tons
- Total Fan Lift: ~1,140 tons
- Total Lift Capacity: ~4,700 tons
- Estimated Structural Weight: ~2,050 tons
- Payload Capacity: ~2,000 tons

Primary Markets

- Mega-infrastructure construction
 - Strategic cargo transport
 - Offshore energy deployment
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Page 9 — Propulsion Layout

The propulsion architecture uses two fan rings.

Outer Ring

Provides directional thrust and yaw control.

Inner Ring

Provides vertical lift and stability.

The distributed propulsion architecture improves safety because loss of a single fan does not compromise flight capability.

Advanced flight control software dynamically balances thrust across the fan network.

Page 10 — Stability and Flight Control

Flight stability is achieved through a combination of buoyant lift and active thrust control.

Control axes include:

- Pitch
- Roll
- Yaw
- Vertical thrust

Sensors continuously monitor wind conditions and structural loads.

The platform maintains stable hovering even during heavy cargo operations.

Page 11 — Industrial Applications

Potential uses include:

- Transport of heavy industrial modules
- Installation of offshore wind turbines
- Mining logistics in remote regions
- Bridge and infrastructure assembly
- Emergency disaster relief cargo transport

These operations are currently limited by the lift capacity of helicopters and the cost of ground logistics.

Page 12 — Strategic Vision

The ATAI L-Series represents a new class of aerial infrastructure platforms.

By combining buoyant lift with distributed propulsion and rigid structural engineering, ATAI platforms can transport cargo masses previously impossible for conventional aircraft.

The scalable architecture allows deployment across multiple industries while maintaining a unified engineering system.

This document outlines the core conceptual engineering framework for the ATAI heavy lift aerial platform family.

Founder

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